

Mechanical Design of a Cognitive Quadruped Robot

1. Introduction

Quadruped robots have become increasingly important in robotics because they can traverse uneven terrain more effectively than wheeled robots while maintaining high stability and maneuverability. These robots are widely used in applications such as search and rescue, industrial inspection, environmental monitoring, and intelligent human assistance.

This report presents the preliminary mechanical design of a **Cognitive Quadruped Robot**, inspired by the Stanford Pupper platform. Unlike conventional robotic dogs that focus only on locomotion, the proposed robot integrates cognitive capabilities through Artificial Intelligence and Computer Vision, enabling it to perceive its environment, recognize obstacles, and make simple navigation decisions.

The objective of this design is to develop a lightweight, stable, and modular robotic platform suitable for future AI integration while maintaining simplicity in manufacturing and assembly.

2. Robot Concept

The proposed robot is designed as a **Cognitive Quadruped Robot** capable of autonomous navigation using onboard sensors and computer vision algorithms. The robot combines mechanical stability with intelligent perception to operate in indoor and semi-structured environments.

Proposed Applications

- Human assistance
- Indoor inspection
- Environmental monitoring
- Obstacle detection
- AI and robotics education

3. Mechanical Design

3.1 Body Shape Selection

Several chassis geometries were evaluated before selecting the final design.

Design	Advantages	Disadvantages
Rectangular	Simple manufacturing, large internal space	Less biomimetic
Elliptical	Better weight distribution, natural appearance	More difficult to manufacture
Modular Frame	Easy maintenance and upgrades	More assembly complexity

The **rectangular chassis with rounded corners** was selected because it offers sufficient internal volume for electronic components while simplifying manufacturing using 3D printing. Rounded corners reduce stress concentration and provide a smoother external appearance.

3.2 Material Selection

Different materials were considered for the robot chassis.

Material	Advantages	Disadvantages
PLA	Easy printing, inexpensive	Brittle
PLA+	Stronger than PLA, lightweight, easy to print	Moderate heat resistance
PETG	Tough and impact resistant	Harder to print
Aluminum	High strength	Heavy and expensive

PLA+ was selected because it offers a good balance between strength, weight, and printability.

3.3 Body Dimensions

The selected dimensions are based on maintaining stability while providing sufficient space for internal components.

Parameter	Value
Length	350 mm
Width	180 mm
Height	90 mm

The dimensions provide enough space for the Raspberry Pi, battery, camera, motor driver, IMU, and wiring while maintaining a compact design.

3.4 Leg Configuration

The robot employs four identical legs arranged symmetrically around the chassis.

Each leg contains three rotational joints:

- Hip Abduction
- Hip Pitch
- Knee Pitch

Therefore,

Parameter	Value
Number of Legs	4
Joints per Leg	3
Total Degrees of Freedom	$4 * 3 = 12$ DoF

The 12-DoF configuration provides sufficient flexibility for stable walking, obstacle avoidance, and future implementation of adaptive gait algorithms.

3.5 Actuator Selection

Several actuator technologies were considered.

Actuator	Advantages	Disadvantages
DC Motor	Simple and inexpensive	Requires external position control
Stepper Motor	Accurate positioning	Heavy and inefficient for walking robots
BLDC Motor	High efficiency	Complex control and expensive
Servo Motor	High positioning accuracy, compact size, easy control	Limited rotation range

Servo motors were selected because they provide precise angular control, sufficient torque for medium-sized quadruped robots, and can be easily controlled using PWM signals generated by the Raspberry Pi.

3.6 Walking Mechanism

The proposed robot uses a **Trotting Gait**, where diagonal leg pairs move simultaneously.

Advantages of this gait include:

- High stability
- Smooth locomotion
- Reduced energy consumption
- Simple gait generation

The robot is expected to maintain continuous ground contact with at least two legs during locomotion, improving balance while reducing vibration.

4. Mechanical Analysis

4.1 Estimated Robot Weight

The total robot weight was estimated based on the masses of the primary mechanical and electronic components. Since this is a preliminary design, the values are approximate and intended for design calculations.

Component	Quantity	Estimated Mass (kg)
PLA+ Chassis	1	0.45
Servo Motors	12	0.66
Raspberry Pi 4	1	0.05
Camera Module	1	0.02
Battery Pack	1	0.25
Motor Driver & Electronics	1	0.12
Wiring & Fasteners	X	0.10
Total Estimated Mass		1.65 kg

The total mass is calculated as:

$$m_{total} = \sum_{i=1} m_i$$

$$m_{total} = 0.45 + 0.66 + 0.05 + 0.02 + 0.25 + 0.12 + 0.10$$

$$m_{total} = 1.65kg$$

Therefore,

Estimated Robot Mass = 1.65 kg

4.2 Robot Weight

The gravitational force acting on the robot is

$$W = mg$$

where

- $m=1.65 \text{ kg}$
- $g=9.81 \text{ m/s}^2$

Therefore,

$$W = 1.65 * 9.81$$

$$W = 16.19 \text{ N}$$

The robot weight is approximately

16.2 N

4.3 Torque Calculation

The hip joint experiences the highest torque because it supports the body during walking.

During a trotting gait, approximately two legs support the robot at one time.

Therefore,

$$F = \frac{W}{2}$$

$$F = \frac{16.19}{2}$$

$$F = 8.10 \text{ N}$$

Assuming the distance from the hip joint to the center of mass of the leg is

$$r = 0.10 \text{ m}$$

The required torque becomes

$$\tau = F \times r$$

$$\tau = 8.10 \times 0.10$$

$$\tau = 0.81 \text{ N} \cdot \text{m}$$

Applying a safety factor of **2**,

$$\tau_{recommended} = 1.62 \text{ N} \cdot \text{m}$$

Therefore, a servo motor capable of producing at least **1.6 N·m** of torque is recommended.

4.4 Walking Speed

The walking speed can be estimated using

$$v = L_s \times f$$

where

- $L_s = \text{stride length}$
- $f = \text{step frequency}$

Assuming

Stride Length = **0.15 m**

Step Frequency = **2 steps/s**

Therefore,

$$v = 0.15 \times 2$$

$$v = 0.30 \text{ m/s}$$

Estimated walking speed:

$$\mathbf{0.3 \text{ m/s}}$$

This speed is suitable for indoor navigation and cognitive perception tasks.

4.5 Center of Gravity

To improve stability, the battery is positioned near the geometric center of the robot body because it is the heaviest component.

The center of gravity is estimated using

$$x_{cg} = \frac{\{\sum m_i x_i\}}{\{\sum m_i\}}$$

And,
$$y_{cg} = \frac{\{\Sigma m_i y_i\}}{\{\Sigma m_i\}}$$

Placing heavy components close to the center minimizes rotational moments and improves balance during locomotion.

4.6 Stability Analysis

The robot employs a **trotting gait**, where diagonal legs move simultaneously while the remaining legs support the body.

This gait provides:

- Good static stability during slow movement.
- Improved dynamic balance.
- Smooth weight transfer.
- Reduced oscillations.

The rectangular body and centrally located battery reduce tipping moments and improve overall stability.

5. Expected Mechanical Challenges

Several mechanical challenges may arise during implementation:

- Increased power consumption due to twelve servo motors.
- Heat generation in actuators during continuous operation.
- Structural deformation if wall thickness is insufficient.
- Battery capacity limiting operating time.
- Reduced walking performance on highly uneven terrain.

These challenges can be minimized by selecting lightweight materials, optimizing gait parameters, and improving structural reinforcement.

References

1. Stanford Robotics Club. *Stanford Pupper: Open-Source Quadruped Robot*.
2. Siciliano, B., & Khatib, O. *Springer Handbook of Robotics*. Springer.
3. Unitree Robotics. *Go2 Robot Specifications*.